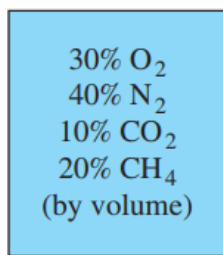
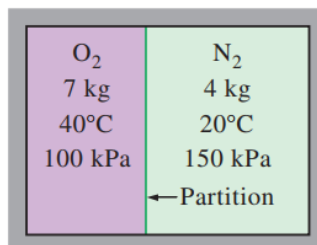


Gas Mixture

1. A gas mixture consists of 4 kg of O_2 , 5 kg of N_2 , and 7 kg of CO_2 . Determine (a) the mass fraction of each component, (b) the mole fraction of each component, and (c) the average molar mass and gas constant of the mixture.
2. The mass fractions of a mixture of gases are 15 percent nitrogen, 5 percent helium, 60 percent methane, and 20 percent ethane. Determine the mole fractions of each constituent, the mixture's apparent molecular weight, the partial pressure of each constituent when the mixture pressure is 1200 kPa, and the apparent specific heats of the mixture when the mixture is at the room temperature.
3. The volumetric analysis of a mixture of gases is 30 percent oxygen, 40 percent nitrogen, 10 percent carbon dioxide, and 20 percent methane. Calculate the apparent specific heats and molecular weight of this mixture of gases.

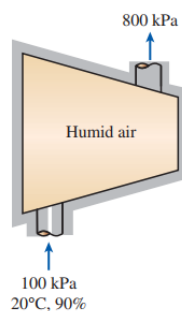


4. An insulated rigid tank is divided into two compartments by a partition. One compartment contains 7 kg of oxygen gas at 40°C and 100 kPa, and the other compartment contains 4 kg of nitrogen gas at 20°C and 150 kPa. Now the partition is removed, and the two gases are allowed to mix. Determine (a) the mixture temperature and (b) the mixture pressure after equilibrium has been established.

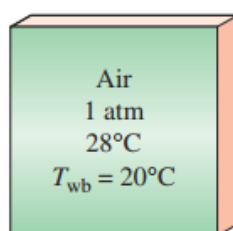


Gas Vapor Mixture

5. Humid air at 100 kPa, 20°C, and 90 percent relative humidity is compressed in a steady-flow, isentropic compressor to 800 kPa. What is the relative humidity of the air at the compressor outlet?



6. The air in a room has a dry-bulb temperature of 26°C and a wet-bulb temperature of 21°C . Assuming a pressure of 100 kPa, determine (a) the specific humidity, (b) the relative humidity, and (c) the dew-point temperature.
7. The air in a room has a pressure of 1 atm, a dry-bulb temperature of 24°C , and a wet-bulb temperature of 17°C . Using the psychrometric chart, determine (a) the specific humidity, (b) the enthalpy, in kJ/kg dry air, (c) the relative humidity, (d) the dew-point temperature, and (e) the specific volume of the air, in m^3/kg dry air.
8. Atmospheric air at a pressure of 1 atm and dry bulb temperature of 28°C has a wet-bulb temperature of 20°C . Using the psychrometric chart, determine (a) the relative humidity, (b) the humidity ratio, (c) the enthalpy, (d) the dew-point temperature, and (e) the water vapor pressure.



9. Air enters a heating section at 95 kPa, 12°C , and 30 percent relative humidity at a rate of $6 \text{ m}^3/\text{min}$, and it leaves at 25°C . Determine (a) the rate of heat transfer in the heating section and (b) the relative humidity of the air at the exit.
10. Air at 1 atm, 15°C , and 60 percent relative humidity is first heated to 20°C in a heating section and then humidified by introducing water vapor. The air leaves the humidifying section at 25°C and 65 percent relative humidity. Determine (a) the amount of steam added to the air, and (b) the amount of heat transfer to the air in the heating section.
11. Air enters a 40-cm-diameter cooling section at 1 atm, 32°C , and 70 percent relative humidity at $120 \text{ m}/\text{min}$. The air is cooled by passing it over a cooling coil through which cold water flows. The water experiences a temperature rise of 6°C . The air leaves the cooling section saturated at 20°C . Determine (a) the rate of heat transfer, (b) the mass flow rate of the water, and (c) the exit velocity of the airstream.
12. Air at 1 atm, 20°C , and 50 percent relative humidity is first heated to 35°C in a heating section and then passed through an evaporative cooler where its temperature drops to 25°C . Determine (a) the exit relative humidity and (b) the amount of water added to air, in $\text{kg H}_2\text{O}/\text{kg dry air}$.
13. A stream of warm air with a dry-bulb temperature of 36°C and a wet-bulb temperature of 30°C is mixed adiabatically with a stream of saturated cool air at 12°C . The dry air mass flow rates of the warm and cool airstreams are 8 and 10 kg/s , respectively. Assuming a total pressure of 1 atm, determine (a) the temperature, (b) the specific humidity, and (c) the relative humidity of the mixture.

Combustion

14. Methane (CH_4) is burned with dry air. The volumetric analysis of the products on a dry basis is 5.20 percent CO_2 , 0.33 percent CO , 11.24 percent O_2 , and 83.23 percent N_2 . Determine (a) the air–fuel ratio and (b) the percentage of theoretical air used.

15. Octane (C_8H_{18}) is burned with dry air. The volumetric analysis of the products on a dry basis is 9.21 percent CO_2 , 0.61 percent CO , 7.06 percent O_2 , and 83.12 percent N_2 . Determine (a) the air-fuel ratio and (b) the percentage of theoretical air used.

Thermodynamic Property Relations (Maxwell Relations)

16. Derive Maxwell Thermodynamic property relation by showing all necessary steps.

17. Verify the validity of the last Maxwell relation ($\left(\frac{\partial s}{\partial p}\right)_T = -\left(\frac{\partial v}{\partial T}\right)_p$) for refrigerant-134a at $50^\circ C$ and 0.7 MPa